

BALGUNTAY, ChinaWMO: **514670**Lat: **42.669N**Lon: **86.329E**Elev: **5687**StdP: **11.92**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	2.6	-19.9	2.3	12.0	-16.1	2.8	12.3	14.0	20.5	12.5	19.7	2.2	90	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	83.6	57.4	81.2	56.6	79.0	56.0	60.7	76.3	59.2	75.1	57.9	74.0	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.8	82.1	64.8	53.8	76.2	63.0	51.9	71.0	63.7	29.7	76.3	28.5	75.1	27.6	74.2	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.6	10.0	DB	-3.3	88.4	4.1	2.3	-6.3	90.0	-8.7	91.3	-11.0	92.6	-14.0	94.2	(4)
(5)			WB	-4.6	62.7	3.8	2.3	-7.4	64.3	-9.6	65.6	-11.7	66.9	-14.5	68.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	18.2	25.9	38.0	50.0	58.3	64.5	67.3	65.9	58.2	46.5	33.9	21.1		
	DBStd	18.27	6.87	6.70	7.34	7.59	6.36	5.38	4.56	4.71	5.48	6.43	6.97	6.66		
	HDD50	3672	985	676	374	91	13	0	0	0	7	147	483	896		
	HDD65	7275	1450	1096	836	451	219	72	27	46	210	574	933	1361		
	CDD50	2122	0	0	3	91	270	436	537	493	254	38	0	0		
	CDD65	249	0	0	0	1	11	58	99	73	7	0	0	0		
	CDH74	2139	0	0	0	17	114	451	839	642	75	1	0	0		
	CDH80	344	0	0	0	1	6	59	173	103	2	0	0	0		
	WSAvg	4.1	4.1	4.8	5.1	5.0	4.5	3.9	3.5	3.5	3.6	3.8	4.3	3.9		
	PrecAvg	7.6	0.0	0.0	0.1	0.2	1.0	1.6	1.8	1.5	1.1	0.2	0.0	0.0		
(15) Precipitation	PrecMax	11.4	0.2	0.2	1.1	0.8	3.0	3.5	5.2	5.6	3.9	0.9	0.1	0.1		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.0	0.0	0.0	0.2	0.2	0.7	1.1	1.3	1.2	0.8	0.2	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	48.4	63.7	76.1	80.6	84.8	88.2	85.9	79.5	70.4	56.7	40.5		
		MCWB	28.6	33.3	42.6	49.9	52.8	56.3	59.7	58.2	55.5	48.7	40.2	29.5		
	2%	DB	35.4	43.7	58.8	71.2	77.3	81.6	84.4	82.8	76.2	65.6	51.8	36.4		
		MCWB	25.7	30.6	40.1	47.5	51.5	55.5	58.4	57.6	53.5	46.5	37.1	26.8		
	5%	DB	31.8	40.3	55.1	67.5	74.2	79.2	81.5	80.4	73.3	62.0	47.9	33.3		
		MCWB	23.4	28.8	38.2	45.6	50.5	54.9	57.6	57.0	52.2	44.6	34.8	24.6		
	10%	DB	28.4	37.0	51.5	64.0	71.3	76.5	78.8	77.8	70.3	58.6	44.4	30.4		
		MCWB	21.0	26.7	36.1	44.2	49.4	54.2	57.2	56.2	51.0	42.6	32.9	22.8		
	0.4%	WB	29.0	33.8	43.3	50.8	54.3	59.2	63.9	63.0	57.4	49.7	40.6	29.7		
		MCDB	38.9	47.8	62.1	73.2	75.8	76.3	77.1	75.9	75.2	68.0	55.5	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.9	31.0	40.6	48.6	52.9	57.6	61.5	60.3	55.0	47.3	37.5	27.1		
		MCDB	35.0	42.9	57.3	68.5	73.9	74.8	76.7	75.5	71.7	63.6	50.6	35.8		
	5%	WB	23.5	29.0	38.6	46.6	51.5	56.5	59.9	58.7	53.5	45.5	35.3	24.9		
		MCDB	31.6	39.6	54.0	65.2	71.6	73.5	75.7	74.6	69.8	60.2	46.8	32.7		
	10%	WB	21.1	27.0	36.5	44.9	50.2	55.5	58.6	57.5	51.9	43.5	33.3	23.0		
		MCDB	28.3	36.5	50.7	62.1	69.3	72.0	74.3	74.0	67.7	57.0	43.8	30.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	21.4	22.0	21.3	20.4	19.9	20.5	21.4	21.8	21.4	18.0	16.3		
		MCDBR	21.4	23.5	26.0	25.5	24.1	23.5	24.8	25.3	25.4	24.8	21.0	17.7		
		MCWBR	14.6	14.8	13.9	11.7	9.3	8.1	8.5	9.3	10.0	11.6	11.9	11.8		
		MCDBR	20.9	22.8	24.6	23.5	21.8	20.6	21.1	21.6	22.9	22.9	20.2	17.4		
	5% WB	MCWBR	14.2	14.4	13.0	10.8	8.7	7.5	8.3	8.9	9.5	10.8	11.6	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.338	0.416	0.482	0.492	0.502	0.468	0.461	0.435	0.379	0.318	0.280		
	taud		2.264	2.108	1.915	1.799	1.834	1.871	2.003	2.003	2.006	2.075	2.196	2.279		
	Ebn at Noon		269	267	257	249	251	248	256	252	248	247	250	259		
	Edh at Noon		28	39	54	65	65	63	54	53	49	40	30	25		
(44) All-Sky Solar Radiation	RadAvg		731	1055	1470	1821	1990	1973	1965	1835	1591	1193	806	618		
	RadStd		42	56	63	104	100	141	102	107	57	62	47	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	-2.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+94	

Nomenclature: See separate page